CONSTRUCTION OS

PRD - Consolidated Domain & Database Architecture

Implementation-ready Product Requirements Document for Product, Enterprise Architecture, Backend, Data Engineering, Security, QA, DevOps, SRE and Platform Operations Teams

Product purpose

Define the governed data model that unifies every ConstructionOS module while preserving one authoritative owner per business fact, explicit company ownership, immutable history, safe cross-domain references and complete traceability.

Document control	Value
Document	PRD_Consolidated_Domain_Database_Architecture
Version	1.0
Status	Implementation-ready platform data architecture and database specification
Audience	Product, enterprise architecture, backend, data engineering, security, QA, DevOps, SRE and technical leadership
Platform scope	All ConstructionOS domains, orchestration services, mobile/offline, client/supplier portals, BIM/Digital Twin, search and analytics
Core architecture	Domain-owned relational records + object storage + typed references + transactional outbox + permission-safe projections + analytical warehouse
Access model	Deny-by-default, tenant isolated, active-company aware, project/record scoped and field restricted
Consistency model	Local ACID transactions within domains; explicit eventual consistency across domains with deterministic reconciliation
Source-of-truth policy	One authoritative owner per business fact; derived stores and projections never become competing masters
Platform Configuration	Mandatory at tenant, group, company, project, module, record type, field, retention and security scope
Date	2026-08-06

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1. Executive Summary

ConstructionOS now contains mature operational domains for projects, units, tasks, documents, CRM, contracts, procurement, inventory, assets, work progress, quality, safety, legal, HR/payroll, finance, calendar, BIM, portals, mobile, reporting, notifications and workflow. The main remaining data risk is not missing functionality; it is allowing those domains to share writable tables, duplicate records or interpret the same business fact differently.

This PRD defines one consolidated domain and database architecture for the complete platform. It assigns one authoritative owner to every business fact, establishes explicit tenant and company ownership, standardizes IDs, versions, temporal fields and data types, and defines how each derived system stores projections without becoming a second source of truth.

Permanent platform rule

One Tenant boundary. Explicit Company ownership. Stable Project context. Independent business domains. No duplicate truth. No hidden database coupling. Complete history and traceability.

1.1 Required outcomes

- Every operational record has a documented authoritative domain and transaction boundary.
- Tenant, Company and Project scope is explicit and validated on every sensitive record and reference.
- Issued, accepted and posted records preserve immutable revisions or corrective transactions.
- Cross-domain relationships use stable typed references, APIs, commands and events.
- Files are stored once and referenced through Documents and exact revisions when required.
- Portal, Mobile, BIM, Search and Reporting stores remain permission-safe, versioned and rebuildable.
- Legal hold, retention, deletion, backup and disaster recovery are governed consistently.
- Database migrations, imports and merges remain auditable, resumable and reconciliation-safe.

1.2 Primary implementation decision

Architecture decision: Use a relational transactional core divided into explicit bounded contexts. Keep local ACID consistency inside each domain, prohibit cross-domain writable foreign-table dependencies, publish committed facts through transactional outboxes, and use versioned projections for derived experiences.

2. Product Vision and Business Outcomes

ConstructionOS should behave as one premium enterprise product while remaining internally divided into governed business domains. Users should navigate connected records naturally, but engineering teams must always know which database owns the record, which version was used, which company is responsible and how a downstream projection can be rebuilt.

Outcome	Required result
Data integrity	No two modules maintain independently editable copies of the same business fact.
Legal and financial control	Company ownership, immutable postings, exact revisions and legal holds remain enforceable.
Scalability	High-volume domains can be separated physically without changing logical contracts.
Security	Tenant, company, project, field and confidentiality scope is applied at storage and service boundaries.
Operational resilience	Events, projections, search and analytics can fail or be rebuilt without corrupting source records.
Implementation efficiency	Shared technical standards reduce repeated schema design and inconsistent lifecycle behavior.
Auditability	Every record, revision, command, event, projection and correction can be traced end to end.

3. Success Metrics and Guardrails

Metric	Initial target	Guardrail
Ownership coverage	100% of aggregate roots assigned to one authoritative domain	No unowned or shared-master records.
Tenant/company completeness	100% of operational roots contain valid ownership scope	No tenant-only shortcut for company-owned data.
Version conflict protection	100% of controlled updates require expected version	No blind last-write-wins on critical records.

Metric	Initial target	Guardrail
Issued-record integrity	0 direct edits to issued/posted records	Corrections create revisions, reversals or adjustments.
Cross-domain coupling	0 direct cross-domain writes	All effects use registered contracts.
Projection traceability	100% of projections carry source ID and version	Derived stores are rebuildable.
Outbox reliability	No committed event permanently lost	Backlog is observable and recoverable.
Backup certification	All Tier 1 domains pass scheduled restore tests	Backup existence alone is insufficient.
Data isolation	0 cross-tenant or unauthorized cross-company disclosure	Counts, search and exports included.
Migration safety	100% production migrations have validation and rollback/compensation	No destructive one-step migration.

4. Scope, Non-Goals and Assumptions

4.1 In scope

- Domain ownership, schemas, aggregate roots and transaction boundaries.
- Tenant, Business Group, Company, Joint Venture, Project, Department, Team and Party structures.
- Canonical IDs, record columns, versions, revisions, temporal fields, statuses and audit.
- Money, currency, quantity, unit, date, timezone, address and contact data standards.
- Documents, files, custom fields, configuration, permissions and confidentiality.
- Domain entity registry covering every completed ConstructionOS module.
- Outbox, inbox, checkpoints, projections, search, analytics and reconciliation storage.
- Indexing, partitioning, constraints, encryption, residency, retention, backup and recovery.
- Migration, import, duplicate matching, merge, data quality, testing and implementation phases.

4.2 Non-goals

- Choosing one final cloud provider or managed database vendor.
- Defining every low-level column for every future feature.
- Replacing individual module PRDs or their domain-specific state machines.
- Implementing external banking, e-signature, identity or file-storage integrations.
- Allowing a central database team to bypass domain APIs and ownership rules.
- Using analytics, BIM, Mobile or Portal stores as authoritative operational databases.

4.3 Assumptions

- PostgreSQL-compatible relational storage is available for the transactional core.
- Object storage, event broker, search engine and analytical warehouse are independent replaceable services.
- Platform authorization can evaluate tenant, company, project, record and field scope.
- Every domain can publish transactional outbox events and expose validation/reconciliation APIs.
- All timestamps are stored in UTC and presented using user/company/project time zones.

5. Architecture Principles

ID	Principle	Required behavior
DB-P01	One owner per fact	Every material business fact has one authoritative domain.
DB-P02	Explicit ownership	Every company-owned record stores owning_company_id.
DB-P03	No shared writable tables	Cross-domain writes and private-schema dependencies are prohibited.
DB-P04	Local consistency	ACID transactions remain inside one bounded context.
DB-P05	Version awareness	Controlled updates require an expected server version.
DB-P06	Immutable control records	Issued and posted records are corrected, never silently overwritten.
DB-P07	Reference, do not copy	Files and cross-domain records are linked by stable identity.
DB-P08	Derived means rebuildable	Search, projections and analytics must support deterministic rebuild.

ID	Principle	Required behavior
DB-P09	Permission before exposure	Derived stores contain only authorized fields and scopes.
DB-P10	Preserve history	Deletion, merge, migration and correction retain complete traceability.
DB-P11	Effective dating	Membership, configuration, compensation and authority preserve effective periods.
DB-P12	Configuration everywhere	Data behavior remains compatible with Platform Configuration.

6. Terminology and Core Concepts

Term	Definition
Bounded context	Business domain with explicit data ownership, rules and contracts.
Aggregate root	Primary entity that controls a local transaction and its child records.
Authoritative record	Operational record that owns the business truth.
Projection	Derived, permission-safe representation of a source record.
Record version	Technical optimistic-concurrency counter incremented on update.
Business revision	Immutable issued version of a controlled business record.
Transaction time	When ConstructionOS recorded the information.
Effective time	When the information applies to the business.
Typed reference	Cross-domain link containing source/target domain, type and stable ID.
Outbox	Local transaction table storing committed events for reliable publication.
Inbox	Consumer table preventing duplicate event effects.
Legal hold	Governed preservation instruction that blocks destructive disposal.

7. Permanent Data Architecture Decision

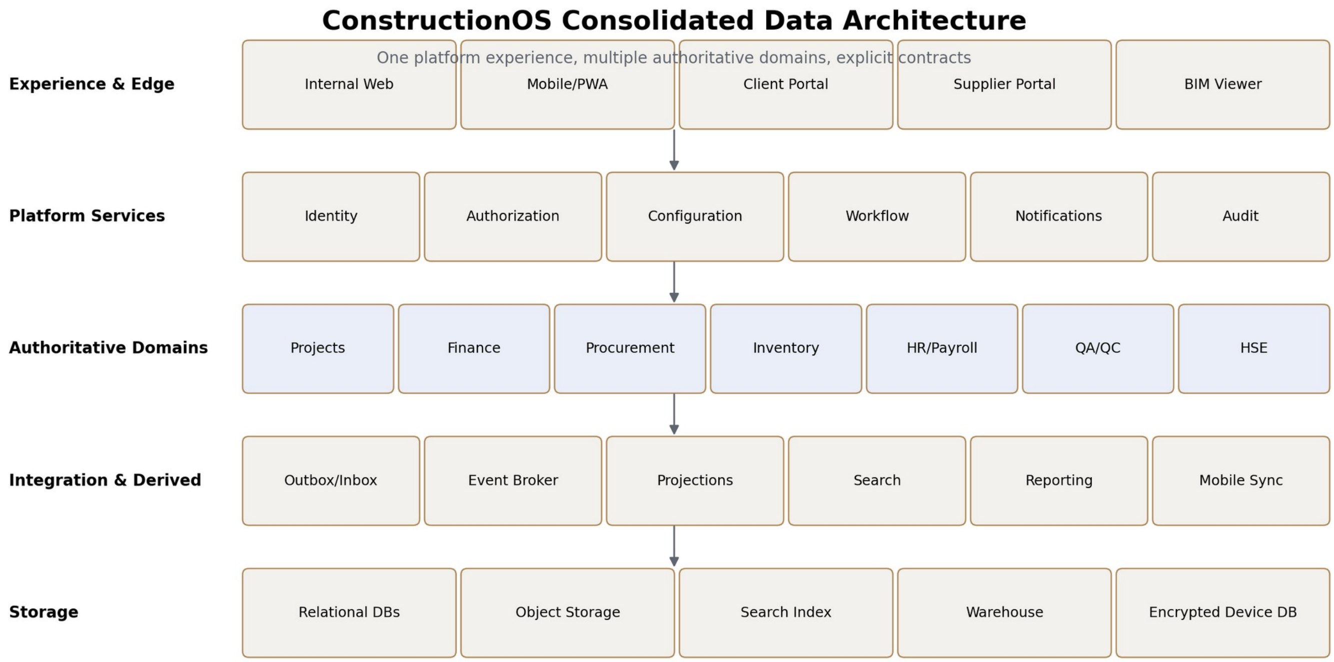
Architecture decision

ConstructionOS will use domain-owned relational records, stable typed cross-domain references, local ACID transactions, transactional outboxes, immutable revisions/postings and permission-safe derived stores. A modular monolith may be used initially, but logical domain boundaries and separate schema ownership are mandatory from the first release.

7.1 Prohibited patterns

- Direct writes into another domain schema.
- Shared master tables edited by several modules.
- Dual writes to two databases inside one request.
- Cascade deletion across domain boundaries.
- Copying source files or business records into related modules.
- Using search indexes or dashboard totals as editable truth.
- Blind mobile database synchronization.
- Portal submissions directly updating internal records.
- Financial calculations using binary floating-point types.
- Hard-deleting posted, issued, accepted or legally held records.

8. Platform Data Reference Architecture



The architecture separates user experiences, platform services, authoritative domains, integration/derived systems and specialized storage. Unified navigation does not imply a shared writable database.

8.1 Data responsibility by layer

Layer	Stores	May not become
Experience & Edge	Transient UI state, secure session context, local drafts	Authoritative business storage.
Platform Services	Identity, permissions, configuration, workflow, delivery and audit	Owner of domain records being orchestrated.
Authoritative Domains	Operational business facts and local histories	Unrestricted shared database.
Integration & Derived	Events, projections, search, sync and analytics	Competing source of truth.
Storage	Technology-specific persistence according to responsibility	A reason to bypass domain boundaries.

9. Storage Technology Strategy

Store	Primary use	Governance
Relational transactional DB	Business records, workflow, permissions and integration state	ACID, constraints, ownership, optimistic concurrency.
Object storage	Files, media, BIM models, exports and upload chunks	Checksum, encryption, retention, legal hold and signed access.
Search index	Global/module search and autocomplete	Derived, minimized, revocation-aware and rebuildable.
Analytical warehouse	Historical facts, dimensions, KPIs and reports	Read-only operational ingestion with reconciliation.
Encrypted device DB	Offline projections, drafts, commands and checkpoints	Expiring, revocable and never authoritative.
Cache	Short-lived query or session acceleration	No durable business truth; invalidation and TTL required.

10. Logical Database Topology

```
ConstructionOS Platform
|-- Foundation: tenant, company, identity, organization, authorization, configuration
|-- Commercial: CRM, units, contracts, procurement, suppliers, portals
|-- Project Delivery: projects, tasks, progress, documents, calendar, QA/QC, HSE, legal, BIM
|-- Resources: inventory, assets, HR/payroll
|-- Financial: finance/accounting
`-- Orchestration: workflow, notifications, mobile sync, search, audit, reporting
```

Each logical domain receives one schema owner, one migration stream, one API/service boundary and one event catalogue. Physical co-location does not permit cross-domain writes.

11. Physical Deployment and Service Evolution

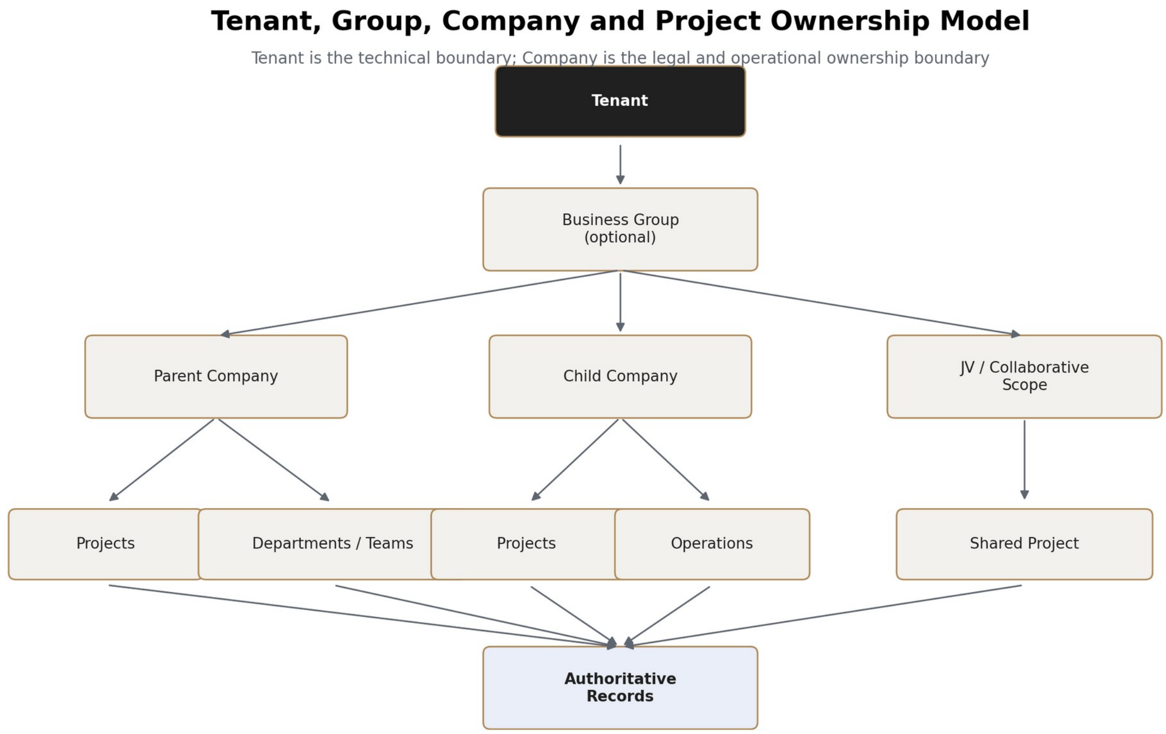
11.1 Initial deployment

- One managed relational cluster may host several domain schemas.
- Each domain uses a separate database role with write access only to its own schema.
- Cross-schema writes are denied at database privilege level.
- Outbox tables live inside the source domain transaction.
- Documents, Workflow, Notifications, Mobile, Search, Reporting and BIM processing may be separate services immediately.

11.2 Separation triggers

Trigger	Reason to separate
Independent scale	High transaction or processing volume requires dedicated capacity.
Security boundary	Payroll, finance or legal data requires stronger isolation.
Availability target	A domain requires different RTO/RPO or maintenance windows.
Storage technology	BIM, search or analytics needs non-relational storage.
Release cadence	A service must deploy independently without schema coupling.
Regional residency	A tenant or domain must remain in a dedicated region.

12. Tenant, Group, Company and JV Hierarchy



12.1 Permanent boundaries

- Tenant is the primary technical isolation boundary.
- Company is the primary legal and operational ownership boundary.
- Business Group organizes related companies but does not erase company isolation.
- Project is a business scope and may include several participating companies.
- Joint Venture access is explicit, effective-dated and project-scoped.
- Every authoritative record still has one owning company, including JV records.

12.2 Foundation entity register

Entity	Key purpose	Key fields
Tenant	Technical isolation and hosting policy	id, code, status, region, locale, timezone, default currency.
BusinessGroup	Optional corporate hierarchy	id, tenant_id, parent_company_id, reporting_currency.
Company	Legal/operational owner	id, group_id, legal identity, jurisdiction, base_currency, status.
CompanyRelationship	Parent/child/shared-service/JV relationship	source_company_id, target_company_id, type, effective dates.
Branch	Operational office or branch	company_id, code, address, timezone.
JointVentureScope	Project collaboration boundary	project_id, lead_company_id, participants, authority scope.

13. Shared Foundation and Master Data

Shared foundation entities may be referenced by all domains, but each has one owning service and controlled mutation API. Shared identity does not imply shared commercial, employment or supplier state.

Master domain	Authoritative entities	Consumers
Foundation	Tenant, BusinessGroup, Company, Branch	All domains.
Organization	Department, Team, Position, Membership	HR, Projects, Tasks, Workflow, Reporting.
Party	Party, Organization identity, Person identity	CRM, Procurement, Contracts, Portals, Legal.

Master domain	Authoritative entities	Consumers
Reference Data	Country, Currency, UOM, Tax code, Classification	All relevant domains.
Platform Configuration	Definitions and effective values	All application and data services.
Authorization	Roles, capabilities, grants and scope policies	Every command, query, projection and export.

14. Internal and External Identity

14.1 Internal separation

Record	Owns	Must not own
UserAccount	Authentication, MFA, sessions, security status	Employment, payroll or visible professional profile.
UserProfile	Visible professional identity and preferences	Private employment or compensation.
EmployeeRecord	Employment, position, manager, payroll linkage	Authentication credentials.
CompanyMembership	Effective relationship to a company	Automatic permission to all company records.
GroupMembership	Group-level role and effective period	Automatic access to every child company.

14.2 External separation

Entity	Purpose
ExternalPerson	Non-employee individual identity.
ExternalOrganization	Client, supplier, subcontractor, consultant or other external legal entity.
PortalAccount	External authentication, sessions, devices and MFA.
PortalMembership	Effective-dated relationship between person and external organization.
BusinessAccessRelationship	Explicit scope to company, project, unit, contract, package or document room.
ExternalSubmission	Immutable external request or response awaiting internal processing.

15. Shared Party Model

A reusable Party identity prevents duplicate legal names and addresses while preserving domain-specific roles. CRM, Procurement, Contracts, External Portals and Legal may all reference one Party, but they retain separate lifecycle and risk records.

```
Party
|-- CRMClientProfile
|-- SupplierProfile
|-- ContractParty
|-- ExternalOrganization
`-- AuthorityProfile
```

Rule	Requirement
Canonical identity	Party stores legal identity and normalized identifiers only.
Domain profiles	Commercial, supplier, client and authority statuses remain in their owning domains.
Duplicate handling	Potential duplicates enter review; no automatic destructive merge.
Merge safety	Issued records and references are migrated transactionally; aliases remain historical.
Privacy	Person Party data follows applicable field and retention policies.

16. Project, Spatial and Work-Breakdown Hierarchies

16.1 Project root

```
Project
|-- ProjectCompanyParticipation
|-- ProjectParticipant / ProjectRoleAssignment
|-- ProjectLocation hierarchy
|-- WorkBreakdownNode hierarchy
`-- Source-linked operational records
```

16.2 Spatial hierarchy

ProjectLocation uses a flexible parent-child model supporting Site, Structure, Building, Level, Zone, Space, Room, Unit and custom configured levels. Infrastructure and industrial projects are not forced to create irrelevant building levels.

16.3 Work breakdown

WorkBreakdownNode is separate from spatial hierarchy and supports Phase, Milestone, Work Package, Trade Package, Cost Package and Activity Group. A work package may span several locations, and a location may contain several work packages.

Integrity rule	Required behavior
No circular paths	Hierarchy updates validate parent paths transactionally.
Stable identity	Moving a node does not change its ID.
Path projection	Materialized path may accelerate reads but is rebuildable.
Effective participation	Company and user participation is effective-dated.
Archive safety	Archived parent nodes preserve historical child links.
Permission scope	Location and work-package scope may restrict access independently.

17. Domain Ownership and Schema Registry

Schema / domain	Primary aggregate roots
foundation	Tenant, BusinessGroup, Company, Branch
identity	UserAccount, UserProfile, Session, Device
organization	Department, Team, Membership, Position
authorization	Role, Capability, Grant, Policy
projects	Project, ProjectLocation, WorkBreakdownNode
units	Unit, Reservation, BuyerRelationship
tasks	Task, Assignment, Checklist, Comment
documents	Document, Revision, Folder, Shortcut
crm	Lead, Opportunity, ClientProfile, Interaction
contracts	Contract, Revision, Obligation, Amendment
procurement	SupplierProfile, Requisition, RFQ, Quotation, PurchaseOrder
inventory	Item, Warehouse, StockMovement, GoodsReceipt
assets	Asset, Assignment, Maintenance, MeterReading
work_progress	Baseline, ProgressEntry, DailyReport, Constraint
quality	Inspection, NCR, Defect, HandoverPackage
hse	Hazard, Incident, PermitToWork, CorrectiveAction
legal	Permit, LegalMatter, Obligation, LegalHold
hr	Employee, Attendance, Timesheet, PayrollRun
finance	Account, Journal, Invoice, Payment, Budget
calendar	CalendarProjection, Meeting, ResourceReservation

Schema / domain	Primary aggregate roots
bim	Model, ModelVersion, Federation, StableObject
workflow	Definition, Instance, WorkItem, Decision
notifications	Policy, Notification, DeliveryAttempt, Acknowledgement
portals	Membership, Projection, Submission, DocumentRoom
mobile	Device, Workspace, OfflineCommand, Conflict
reporting	MetricDefinition, Dataset, Snapshot, Report
audit	AuditEvent, SensitiveAccessEvent
integration	Outbox, Inbox, Checkpoint, ReconciliationRun

18. Canonical Record Contract

```
id                      UUID / time-ordered UUID
tenant_id               UUID
owning_company_id       UUID, where applicable
project_id              UUID, where applicable
record_version          BIGINT
lifecycle_status        domain-controlled value
confidentiality_class   controlled value
created_at / created_by
updated_at / updated_by
archived_at / archived_by, nullable
source_system / import_batch_id, nullable
correlation_id, nullable
```

Requirement ID	Requirement
DB-REC-001	Every aggregate root contains tenant_id and immutable id.
DB-REC-002	Every company-owned aggregate contains owning_company_id.
DB-REC-003	Project-scoped records store project_id even when inferable from another reference.
DB-REC-004	record_version increments on every controlled update.
DB-REC-005	Created/updated actor and UTC timestamps are mandatory.
DB-REC-006	Technical archive state remains separate from business lifecycle status.
DB-REC-007	Confidentiality classification is available on sensitive roots and inherited safely.
DB-REC-008	Import and correlation provenance remains traceable.

19. Identifier and Human-Code Standards

Identifier type	Standard
Primary key	Globally unique immutable UUID; time-ordered variant preferred for high-volume roots.
Human code	Separate readable code, unique within configured scope and never used as primary key.
External identifier	Source-system identifier stored in an effective-dated mapping table.
Document number	Controlled sequence independent from file name.
Event/command ID	UUID with idempotency and correlation context.
Alias	Historical code or merged identity preserved for lookup and audit.

Code rule

Changing a display code must not break references. Stable internal IDs are permanent; old codes remain aliases when renumbering is authorized.

20. Aggregate Roots and Transaction Boundaries

Only an aggregate root may control mutation of its child records. Cross-domain references are validated through contracts but are not part of the local transaction.

Aggregate	Local children	Explicitly external
PurchaseOrder	Lines, delivery schedules, commercial conditions, revision metadata	Inventory receipt, Finance invoice/payment, Contract record.
Inspection	Checklist results, measurements, participants, findings, evidence references	Task execution, Inventory quantity, Procurement supplier record.
JournalEntry	Journal lines, dimensions, source references	Contract, payroll or invoice source record.
Task	Assignments, checklist, comments, dependencies, evidence links	Inspection, workflow decision or source business status.
WorkflowInstance	Stages, nodes, work items, decisions, timers	Source record and final business transition.
Document	Revisions, file object links, issue state	Records linking to the document.
ProgressEntry	Quantities, crews, equipment/material usage, evidence references	Quality acceptance and Finance certification.

21. Cross-Domain Reference Contract

```
DomainReference
- tenant_id
- source_domain / source_entity_type / source_entity_id / source_version
- target_domain / target_entity_type / target_entity_id
- relationship_type
- created_at / created_by / status
```

ID	Requirement
DB-REF-001	Creating a cross-domain link validates tenant and current target existence through the owning contract.
DB-REF-002	Exact target version is stored when business meaning depends on a revision.
DB-REF-003	No cross-domain cascade delete is permitted.
DB-REF-004	Orphaned or stale references are detected through reconciliation.
DB-REF-005	Viewing the source does not grant permission to open the target.
DB-REF-006	References remain historical after target archive or supersession.
DB-REF-007	Polymorphic reference types are registry-controlled, not arbitrary strings.

22. Record Versioning and Business Revisions

22.1 Optimistic concurrency

```
UPDATE aggregate
SET ..., record_version = record_version + 1
WHERE id = :id
  AND tenant_id = :tenant_id
  AND record_version = :expected_version
```

No updated row returns a version-conflict result. The application shows the current server version and preserves the user draft when possible.

22.2 Business revision

Technical version	Business revision
Increments on every committed update.	Created only when a controlled/issued record changes.
Supports concurrency.	Preserves externally or operationally meaningful versions.

Technical version	Business revision
May change while a draft is edited.	Published revisions are immutable.
Not normally shown as a document revision label.	Uses revision sequence/code such as Rev 2 or C.

22.3 Material-change dependencies

- Workflow approval references the exact source version.
- Portal publication references the exact published version.
- Document links may require the exact revision.
- BIM federations reference immutable model versions.
- Mobile commands carry the expected server version.
- Analytics preserves source version or transaction lineage.

23. Temporal Data and Effective Dating

Time concept	Fields	Use cases
Transaction time	created_at, updated_at, recorded_at	When ConstructionOS stored the fact.
Effective time	effective_from, effective_to	When membership, term, permission or configuration applies.
Business event time	submitted_at, issued_at, posted_at, accepted_at	Domain lifecycle milestones.
Device time	device_timestamp, timezone, server_offset	Offline evidence and clock-integrity review.
Bitemporal history	transaction period + effective period	Compensation, authority, terms, assignments and configurations.

Temporal rule

Changing the effective date of a controlled record must preserve when the change was entered and which previously published calculations or decisions used the former value.

24. Lifecycle Status and Immutability

24.1 Status separation

Domain lifecycle statuses remain domain-specific. Shared technical status such as Active, Archived, DeletedPending or ProcessingFailed must not replace the domain state machine.

24.2 Immutability levels

Level	Examples	Correction method
Draft mutable	Draft task, RFQ, contract, inspection	Version-controlled edit.
Submitted controlled	Submitted requisition, claim or inspection request	Return, withdraw, resubmit or new revision.
Issued immutable	Issued PO, published document, final minutes	Superseding revision or addendum.
Posted immutable	Journal, payment, stock movement, locked payroll	Reversal, adjustment, credit/debit or corrective transaction.
Accepted immutable	Inspection acceptance, handover acceptance	Reinspection, reopening or superseding acceptance with reason.

25. Deletion, Archive, Recycle Bin and Legal Hold

25.1 Recycle Bin lifecycle

```
Active -> DeletedPending -> Recycle Bin -> Restored
                                  `-> Retention Expired -> Deletion Review -> Purged
```

25.2 Deletion eligibility

- Only draft or explicitly disposable records may enter Recycle Bin.
- No immutable dependent, posted transaction or issued revision may depend on the record.
- No active legal hold may apply.
- Purge requires retention eligibility, authority, reason and deletion manifest.
- Backups expire according to policy; production deletion does not rewrite immutable backup history immediately.

25.3 Legal hold

Rule	Requirement
Scope	Company, project, user, date range, record type, specific record, document or legal matter.
Deletion	Permanently blocked while hold is active.
Archive	Allowed only when records remain searchable, linked and preserved.
Retention	Normal disposal is paused or recalculated.
Release	Authorized action, reason, evidence and audit required.
Enforcement	Documents enforces files; source domains enforce records.

26. Audit and Activity Architecture

26.1 AuditEvent

```
AuditEvent
- tenant/company/project scope
- domain, entity_type, entity_id, entity_version
- event_type, actor, acting_on_behalf_of
- role/permission snapshot, device/session
- occurred_at, effective_at
- before_data, after_data, reason
- correlation_id, causation_id, classification
```

26.2 Separate histories

History	Purpose
Business audit	Authoritative record changes and corrections.
Workflow audit	Routing, decisions, delegation and finalization.
Integration audit	Event publication, consumption, retries and projection updates.
Security audit	Permission denial, sensitive read, impersonation and export.
Activity timeline	Permission-safe user-facing summary derived from source histories.
Support audit	Temporary company-approved support access and actions.

Immutability rule

Audit records are append-only. Corrections create new audit events; no administrator may edit or permanently delete committed audit history through ordinary product interfaces.

27. Money, Currency, Quantity and Unit Standards

27.1 Money

Field type	Recommended type / rule
Monetary amount	NUMERIC(24,6) or domain-approved greater precision.

Field type	Recommended type / rule
Unit price	NUMERIC(24,8).
Exchange rate	NUMERIC(30,12).
Percentage	NUMERIC(12,8).
Currency	ISO currency code stored with every amount.
Prohibited	FLOAT/DOUBLE for monetary calculations.

27.2 Quantities and UOM

- Every quantity stores numeric value plus unit_id.
- Conversions preserve original value, original unit, converted value, rule version and rounding method.
- Model-derived, ordered, delivered, accepted, certified and paid quantities remain distinct.
- Units include dimension type, base unit, conversion factor and decimal scale.
- Custom units require governed definitions and cannot silently reuse existing symbols.

27.3 Financial state separation

Budget, forecast, commitment, accrual, invoice, certified value, payment, retention and tax are separate records or dimensions. A generic editable cost total is prohibited.

28. Date, Timezone, Address and Contact Standards

Area	Standard
Timestamps	Store UTC TIMESTAMPTZ; present using IANA timezone.
Date-only values	Use DATE for contractual, due and effective dates without time.
Local clock values	Store TIME plus explicit timezone/calendar context.
Address	Structured country, administrative area, city, postal code and optional geospatial coordinates.
Contact point	Typed, normalized, effective-dated email/phone/address relationship.
Sensitive values	Encrypt/tokenize bank, government, payroll and restricted contact fields as required.

29. Documents and File Reference Architecture

```
Document (logical record)
`-- DocumentRevision (immutable issued/file revision)
    `-- FileObject (object storage key, checksum, size, MIME, encryption)

DocumentLink -> target domain/entity and optional exact revision
```

ID	Requirement
DB-DOC-001	File bytes are stored once; other domains create references or shortcuts.
DB-DOC-002	Exact document_revision_id is mandatory when a process depends on a controlled version.
DB-DOC-003	Checksums, malware state, size, MIME type and object key are stored in Documents.
DB-DOC-004	Superseded revisions remain historical and resolvable.
DB-DOC-005	File access is reauthorized through Documents at open/download time.
DB-DOC-006	Legal hold and retention apply to both metadata and file object.
DB-DOC-007	Temporary upload/quarantine objects are not published Document revisions.

30. Custom Fields and Platform Configuration

30.1 Custom fields

ConstructionOS supports controlled extension fields without converting the platform into an ungoverned Entity-Attribute-Value database.

Control	Requirement
Definition	Field key, type, label, validation, visibility, permissions, effective dates and version.
Storage	Typed JSON for low-volume extensions; indexed/generated columns for query-critical fields.
Critical fields	Core ownership, money, lifecycle and security fields remain first-class relational columns.
Type change	Requires migration and validation when values already exist.
Execution	No arbitrary executable code or hidden integration behavior.
Permissions	Field-level rules apply to query, export, search, portal and mobile projections.

30.2 Configuration precedence

```
System Default -> Tenant -> Business Group -> Company -> Project -> Module -> Record Type -> Controlled Scope
```

Every effective configuration value records source scope, inherited value, effective dates, actor, reason and version. Local overrides never destroy the parent configuration history.

31. Authorization and Data Scope

ConstructionOS combines capability-based RBAC with contextual ABAC. Membership, title or administrator status alone does not grant business-record authority.

Dimension	Examples
Principal	User, Team, Company role, External membership, Service identity.
Capability	View, create, edit, submit, approve, post, export, configure.
Scope	Tenant, company, department, project, location, contract, unit, supplier, record.
Restriction	Field policy, confidentiality, value threshold, effective period.
Context	Active company, device trust, delegation, workflow state, source relationship.
Decision log	Policy version, resolved grants, exclusions and reason.

Deny-by-default

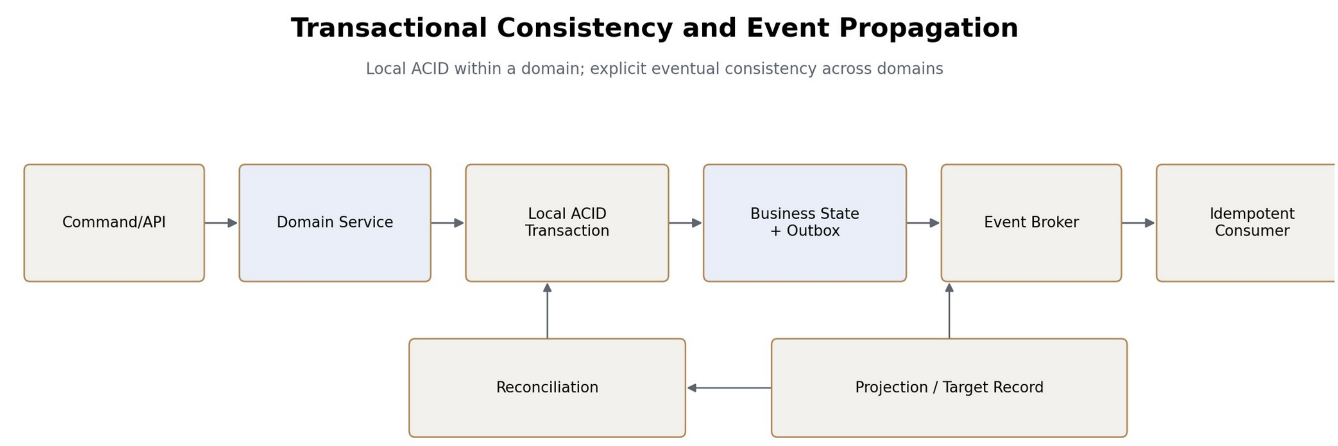
Absence of an explicit effective grant means access is denied. A visible linked summary never grants access to the authoritative target record.

32. Domain Entity Registry

The following registry defines primary aggregate roots and permanent ownership boundaries. Detailed fields and workflows remain in the individual module PRDs.

Domain	Primary aggregate roots	Permanent boundary
Projects	Project, ProjectLocation, ProjectCompanyParticipation, ProjectParticipant, WorkBreakdownNode	Owns project identity and hierarchy; not progress, contracts or finance.
Units	Unit, UnitType, Reservation, BuyerRelationship, UnitVariationRequest	Owns unit commercial lifecycle; QA owns handover readiness; Finance owns money.
Tasks	Task, Assignment, Checklist, Comment, Dependency, TimeEntry	Owns accountable execution; does not constitute formal approval or quality acceptance.
CRM	Lead, Opportunity, ClientProfile, Interaction, Proposal	Owns prospect/client commercial context; Contracts and Finance remain authoritative.
Contracts	Contract, Revision, Party, Obligation, Amendment, Notice, Guarantee	Owns legal agreements and obligations; issued revisions are immutable.
Procurement	SupplierProfile, Requisition, RFQ, Quotation, Evaluation, PO, DeliveryNotice	Owns sourcing and purchasing; Inventory owns stock and Finance owns accounting.
Inventory	Item, Warehouse, StockLot, Balance, Reservation, Movement, Receipt, Count	Owns physical quantity and location; QA owns acceptance.
Assets	Asset, Component, Assignment, Meter, MaintenancePlan, WorkRecord, Disposal	Owns long-lived operational asset lifecycle.
Work Progress	WorkPackage, Baseline, Activity, ProgressEntry, DailyReport, Constraint, Delay	Owns physical execution and quantities; QA owns acceptance.
QA/QC & Handover	QualityPlan, Inspection, NCR, Defect, Commissioning, HandoverPackage, Warranty	Owns quality decisions, readiness and acceptance.
HSE	Hazard, RiskAssessment, Incident, PermitToWork, StopWork, CorrectiveAction	Owns safety events, restrictions and investigations.
Government & Legal	Permit, RegulatoryRequirement, LegalMatter, Submission, Obligation, LegalHold	Owns formal regulatory/legal records and holds.
HR & Payroll	Employee, PositionAssignment, Attendance, Timesheet, PayrollRun, PayrollResult	Owns employment and payroll; Finance owns posting/payment.
Finance	Account, Journal, Budget, Commitment, Invoice, Payment, Reconciliation	Owns accounting and financial truth; posted entries are immutable.
Calendar & Meetings	CalendarProjection, Meeting, Agenda, Decision, MinutesRevision, Reservation	Owns calendar/meeting records; source modules own source dates.
BIM & Digital Twin	Model, ModelVersion, Federation, StableObject, Viewpoint, OperationalLink	Owns model/object identity and visual relationships; not operational states.
Documents	Document, Revision, FileObject, Folder, Shortcut, DocumentLink	Owns file identity, revisions, access, retention and legal hold.
Workflow	Definition, Version, Instance, WorkItem, Decision, Delegation, Timer	Owns decision process; source module owns record and consequence.
Notifications	Policy, Template, Notification, Recipient, DeliveryAttempt, Acknowledgement	Owns communication delivery; source module records what happened.
Portals	External identity, Membership, Relationship, Projection, Submission, DocumentRoom	Owns external access and submissions; source modules own business truth.
Mobile	Device, Workspace, Command, Conflict, Upload, Checkpoint	Owns offline delivery and sync; operational modules own final records.
Reporting	Metric, Dataset, Snapshot, Dashboard, ReportExecution	Owns governed metrics and issued results; operational modules own facts.

33. Integration Persistence



33.1 Required entities

Entity	Purpose	Key control
OutboxMessage	Committed domain event awaiting publication	Written in same transaction as business state.
InboxMessage	Consumer duplicate prevention	Unique consumer_name + event_id.
ConsumerCheckpoint	Partition/stream progress	Resumable ordered processing.
CommandLog	Idempotent command result and provenance	Stable idempotency key.
ReconciliationRun	Source-target correctness comparison	No silent authoritative auto-fix.
DeadLetterRecord	Unresolved event/command failure	Audited replay eligibility and operator action.

33.2 Delivery rules

- Use at-least-once event delivery; exactly-once transport is not assumed.
- Consumers are idempotent and preserve aggregate version ordering where required.
- Late or stale events are ignored, applied historically or routed to reconciliation according to contract.
- Projection rebuild is permitted; rewriting authoritative records requires a source-domain command.

34. Projection and Read-Model Architecture

Projection class	Examples	Freshness/permission requirement
Embedded UI summary	Project finance, supplier contract, unit handover status	Source version and permission-safe fields.
Calendar projection	Task, permit, contract, inspection and delivery dates	Source date ownership preserved.
Portal projection	Client/supplier external views	Allowlisted fields, publication version and relationship scope.
Mobile projection	Offline assigned records and templates	Device/user scope, expiry and revocation.
BIM projection	Progress, quality, HSE, asset and unit overlays	Stable object link and target permission check.
Group dashboard	Consolidated company summaries	Permission-safe incremental read model.
Analytical fact	Historical transaction or state change	Traceable lineage and reconciliation.

34.1 Standard fields

```
projection_id, tenant_id, company/project scope
source_domain, source_entity_type, source_entity_id, source_entity_version
projection_version, generated_at, last_verified_at
freshness_status, access_policy_version, payload
```

35. Search Index Architecture

ID	Requirement
DB-SRCH-001	Search indexes are derived and fully rebuildable.
DB-SRCH-002	Tenant and scope filters apply before result scoring and counts.
DB-SRCH-003	Restricted fields are excluded from indexed payload.
DB-SRCH-004	Permission revocation removes or invalidates indexed records within configured SLA.
DB-SRCH-005	Opening a result rechecks source permission.
DB-SRCH-006	Autocomplete and facets must not reveal hidden record existence.
DB-SRCH-007	Every search document carries source version and index version.

36. Reporting Warehouse Architecture

```
Raw Ingestion -> Validated Staging -> Conformed Dimensions -> Domain Facts -> Semantic Metrics -> Snapshots ->
Dashboards/Reports
```

36.1 Conformed dimensions

- Tenant and Business Group
- Company and Project
- Project Location and Work Package
- Time and Currency
- Party, Client and Supplier
- Contract and Cost Code
- Employee under restricted policy
- Asset and Inventory Item

36.2 Fact families

- Finance transactions
- Procurement spend and delivery
- Inventory movements
- Progress quantities
- Quality inspections/NCR/defects
- HSE events
- Workforce time and payroll
- Asset utilization
- Contract obligations
- Unit sales
- Portal submissions
- Workflow decisions

Permission-before-aggregation

Restricted rows and fields are excluded before totals are calculated. A dashboard or shared report never grants access to data the viewer could not otherwise access.

37. Workflow, Notification, Portal, Mobile and BIM Data

Platform	Persistent records	Permanent boundary
Workflow	Definitions, versions, instances, work items, decisions, timers, locks	References exact source version; does not own source business record.
Notifications	Policies, templates, notifications, recipients, delivery attempts, read/ack state	Does not own task, deadline or approval truth.
Portals	External identities, memberships, relationships, projections, submissions, document rooms	External actions are immutable submissions, not internal edits.
Mobile	Devices, workspaces, manifests, commands, uploads, conflicts, checkpoints	Offline database stores projections/drafts only.
BIM	Models, versions, federations, stable objects, mappings, viewpoints, links	Operational state remains in source modules.
Reporting	Metrics, datasets, snapshots, executions and issued results	Operational facts remain in source modules.

38. Indexing, Partitioning and Constraints

38.1 Baseline indexes

```
(tenant_id, id)
(tenant_id, owning_company_id)
(tenant_id, project_id)
(tenant_id, lifecycle_status)
(tenant_id, created_at)
(source_domain, source_entity_type, source_entity_id)
(idempotency_key) UNIQUE where required
```

38.2 Partition candidates

- Audit and activity events
- Outbox/inbox records
- Notification delivery attempts
- Mobile commands and uploads
- Stock movements
- Journal lines
- Attendance events
- Progress entries
- Analytical staging data

38.3 Database-enforced constraints

- Tenant/company ownership consistency and valid effective-date ranges.
- Unique active codes, memberships, idempotency keys and published revisions.
- No circular project, location, department or work-breakdown paths.
- Balanced Finance journals and controlled fiscal-period states.
- One active project baseline and one current published revision.
- Valid currency, UOM and source-reference registry values.
- No overlapping active configuration or authority ranges where prohibited.

39. Multi-Company and Joint Venture Integrity

Rule	Required behavior
Tenant consistency	All foreign IDs and typed references belong to the same tenant.
Active company	Sensitive commands require explicit active_company_id.
Group isolation	Group membership does not create child-company access.
Project participation	Cross-company project records require active participation and scope.

Rule	Required behavior
JV ownership	Each authoritative record retains one owning company.
Consolidation	Group totals use permission-safe read models, not live unrestricted joins.
Shared staff	Employment remains company-owned; project/team assignments create scoped participation.

40. Classification, Encryption and Residency

40.1 Classification

Recommended classes: Public, External Shared, Internal, Commercial Restricted, Financial Restricted, HR Confidential, Payroll Restricted, HSE Sensitive, Legal Restricted, Executive Restricted and System Secret.

40.2 Encryption

- Database volumes, object storage, backups and device databases are encrypted at rest.
- Bank details, government identifiers, payroll, medical/restricted HR and secrets use field-level encryption or tokenization.
- Keys are rotated and isolated by environment; dedicated tenants may use tenant-specific keys.
- All client and service connections use authenticated encrypted transport.
- No secrets or unrestricted personal values are written to logs or event payloads.

40.3 Residency

Tenant field	Purpose
home_region	Primary processing and authoritative storage region.
permitted_processing_regions	Regions allowed for services and workers.
permitted_backup_regions	Backup and disaster-recovery locations.
cross_region_replication_policy	Allowed replication and failover behavior.
data_residency_policy	Domain or classification-specific restrictions.

41. Retention, Backup and Disaster Recovery

41.1 Retention precedence

```
Legal Hold -> Statutory Retention -> Contractual Retention -> Company Policy -> User Deletion Request
```

41.2 Recovery tiers

Tier	Domains	Protection
Tier 1 - Critical	Identity, Authorization, Finance, Documents metadata, Workflow, HR/Payroll	Continuous logs, PITR, cross-zone copy, frequent restore tests.
Tier 2 - Operational	Projects, Tasks, Procurement, Inventory, Assets, Progress, QA/QC, HSE, Legal	PITR, scheduled backups and tested domain restore.
Tier 3 - Rebuildable derived	Search, projections, mobile packages, BIM artifacts, analytical aggregates	Source replay/rebuild plus metadata backups.

41.3 Backup controls

- Encrypted full and incremental backups with separate credentials.
- Point-in-time recovery and object-storage versioning.
- Cross-zone or approved cross-region replication.
- Ransomware-resistant protected copy.
- Restore validation against integrity checks and application smoke tests.
- Backup access and restore actions are audited.

42. Migration, Import, Duplicate and Merge

42.1 Schema migration

```
Add new structure -> dual write/read compatibility -> backfill -> validate -> switch reads -> stop old writes
-> remove later
```

- Every production migration is versioned, checksum-protected and tenant-aware.
- Backfills run in resumable batches with checkpoints and reconciliation.
- Destructive changes require a later release after compatibility verification.
- Rollback or compensating action is documented before deployment.

42.2 Imports

```
Upload -> Quarantine -> Parse -> Staging -> Validate -> Duplicate Match -> Preview -> Authorized Commit ->
Reconcile -> Report
```

42.3 Merge

- Select one surviving master and preserve aliases.
- Migrate references transactionally and preserve issued histories.
- Block unsafe merges involving active commercial or legal records until review.
- Publish merge events and rebuild derived projections.
- Never delete the historical identity without a merge audit record.

43. Data Quality and Reconciliation

Quality check	Examples
Ownership	Missing tenant/company/project or invalid participation.
Reference integrity	Broken or stale typed references, missing exact revisions.
Financial integrity	Unbalanced journals, currency mismatch, posting divergence.
Quantity integrity	Ordered/received/accepted/certified variance.
Projection integrity	Missing, stale or permission-mismatched derived records.
Identity integrity	Potential duplicate Party, supplier, client or employee.
Temporal integrity	Overlapping effective records or invalid date ranges.
Security integrity	Restricted fields indexed, cached or published incorrectly.

Data-quality tooling may identify and route issues or safely rebuild derived projections. It may not silently change authoritative business records.

44. Observability and Operational Support

Signal	Purpose
Connection/query latency	Detect capacity and query regressions.
Lock contention/deadlocks	Detect transaction-boundary problems.
Replication lag	Protect read consistency and failover readiness.
Table/index growth	Plan partitions and capacity.
Outbox backlog	Detect committed facts not published.
Consumer lag/failure	Detect delayed downstream state.
Projection freshness	Detect stale Portal, Mobile, BIM, Search or Calendar data.
Backup age/restore status	Prove recoverability.

Signal	Purpose
Migration progress	Detect stalled or partially applied schema changes.
Data-quality mismatches	Expose source-target drift and isolation failures.

Operational requirement

Every critical domain, consumer, reconciliation job, backup process and migration has an owner, dashboard, alert threshold and recovery runbook.

45. Non-Functional Requirements

Area	Requirement
Availability	Tier 1 data services use multi-zone resilience and tested failover.
Consistency	Local aggregate transactions are atomic; cross-domain state is explicit and reconcilable.
Scalability	Large tenants and high-volume tables support partitioning and dedicated deployment.
Performance	Indexed operational list queries meet agreed p95 targets without unrestricted cross-domain joins.
Security	Tenant, company, project and field isolation is enforced at service and storage layers.
Recoverability	RPO/RTO are defined by tier and verified through restore exercises.
Maintainability	Schema ownership, migrations, contracts and data dictionaries are versioned.
Accessibility	Data architecture supports localized labels, formats and assistive frontend behavior.
Auditability	Critical changes and sensitive reads remain reconstructable.
Portability	External vendors remain replaceable; authoritative IDs and contracts are platform-owned.

46. Testing and Certification Strategy

46.1 Required test categories

- Schema and migration tests
- Aggregate transaction tests
- Constraint and version-conflict tests
- Tenant/company/project isolation tests
- Cross-domain contract tests
- Outbox/inbox idempotency tests
- Projection rebuild and revocation tests
- Financial precision and balancing tests
- Document revision and checksum tests
- Legal hold and retention tests
- Backup restore and disaster recovery tests
- Import/merge/reconciliation tests
- Search leakage tests
- Mobile/Portal/BIM boundary tests
- Performance and partition tests

46.2 Critical scenarios

1. A company user attempts to reference a sibling-company record through a shared group project.
2. Two devices update the same controlled aggregate version.
3. A posted payment correction is attempted through direct edit.
4. A document revision used by an approved workflow is superseded.
5. A portal relationship is revoked while its projection remains cached.
6. A mobile command is delivered three times after a timeout.
7. A supplier duplicate merge has active Purchase Orders and contracts.

8. An analytical rebuild receives late events and changed company assignments.
9. A legal hold is applied after a record enters Recycle Bin but before purge.
10. A database migration stops halfway through a multi-million-row backfill.
11. A backup restore is performed without current search/analytics indexes.
12. A BIM model revision splits one stable object into several objects.

47. Acceptance Criteria

ID	Area	Acceptance criterion
AC-001	Ownership	Every aggregate root has one documented authoritative domain.
AC-002	Tenant isolation	No record, query, event, index or export crosses tenant boundaries.
AC-003	Company ownership	All company-owned roots contain and validate owning_company_id.
AC-004	Project scope	Project references validate company participation and effective access.
AC-005	Concurrency	Controlled updates reject stale expected versions.
AC-006	Revisions	Issued records change only through controlled new revisions/addenda.
AC-007	Posted records	Financial and stock postings cannot be edited or hard-deleted.
AC-008	References	Cross-domain links use registered typed references and no cascade delete.
AC-009	Documents	Files are stored once; exact revisions remain historical and resolvable.
AC-010	Outbox	Committed domain events originate from the local transactional outbox.
AC-011	Idempotency	Duplicate commands/events do not duplicate business effects.
AC-012	Projections	Every derived record carries source ID/version and can be rebuilt.
AC-013	Search	Hidden records are not inferable through results, counts or autocomplete.
AC-014	Portal	External writes remain immutable submissions until internal processing.
AC-015	Mobile	Offline mutations are commands; device storage never becomes authoritative.
AC-016	Workflow	Decisions reference exact source version and completion requires source finalization.
AC-017	Analytics	Official metrics reconcile to operational sources and apply permissions before aggregation.
AC-018	Legal hold	Active holds block destructive disposal across files and records.
AC-019	Retention	Purge follows policy precedence and retains an auditable deletion manifest.
AC-020	Recovery	Tier 1 and Tier 2 domains pass documented restore tests.
AC-021	Migrations	Production schema changes use expand/contract and validation.
AC-022	Merge	Duplicate merges preserve aliases, references, audit and issued records.
AC-023	Data quality	Reconciliation detects missing, stale, mismatched and orphaned records.
AC-024	Observability	Critical database and integration signals have alerts, owners and runbooks.
AC-025	Configuration	Custom fields and effective data behavior honor Platform Configuration.

48. Implementation Phases

Phase	Scope	Exit condition
1 - Data foundation	Tenant/group/company, canonical IDs, ownership fields, common columns, authorization and audit contracts.	All new aggregates follow common record contract.
2 - Domain separation	Schema ownership, aggregate services, typed references, optimistic versions and constraints.	No direct cross-domain writes remain.
3 - Integration persistence	Outbox, inbox, checkpoints, projection registry and reconciliation.	Reference vertical flow is retry-safe and traceable.
4 - Record governance	Business revisions, immutability, corrections, archive, Recycle Bin, legal hold and retention.	Critical records pass governance tests.
5 - Derived platforms	Search, Portal, Mobile, BIM and Reporting projection contracts.	Every derived store is permission-safe and rebuildable.
6 - Scale and resilience	Partitioning, dedicated services, enterprise tenant isolation, regional residency and DR.	Capacity and recovery certification approved.

48.1 Reference implementation flow

Use one complete vertical path as the reference implementation before broad migration: Contract Approved -> Finance Structure -> Invoice -> Payment Approval -> Payment Posting -> Portal/Reporting/Notification projections. The pattern must demonstrate ownership, commands, outbox, versioning, permission filtering, reconciliation and audit.

49. Risks and Open Decisions

Topic	Decision required / risk
Physical isolation	Which domains receive dedicated databases in the first production release.
UUID strategy	Confirm time-ordered UUID format and database/library support.
Row-level security	Use as mandatory enforcement or defence-in-depth only.
Schema tooling	Select migration framework, contract generation and data dictionary tooling.
Custom fields	Define indexing thresholds and supported query operators.
Field encryption	Confirm key hierarchy, searchable encryption limitations and rotation.
Residency	Confirm supported regions and dedicated-tenant options.
Retention	Approve record-by-record statutory and contractual retention matrix.
Audit storage	Determine immutable/WORM storage requirements and retention tiers.
Warehouse technology	Select analytical store and CDC/event ingestion posture.
Search technology	Confirm security-filter strategy and revocation SLA.
Legacy migration	Map existing duplicated records, files, codes and ownership gaps.
Data stewardship	Assign domain owners and data stewards for every schema.
RPO/RTO	Approve service-tier targets and restore-test frequency.

50. Definition of Done and Required Deliverables

50.1 Definition of Done

- Every aggregate root and authoritative fact is registered with one owning domain.
- Canonical record, ID, version, temporal, money, quantity and audit standards are implemented.
- Tenant, company, project and confidentiality isolation tests pass.
- No production cross-domain direct writes or hidden writable shared tables remain.
- Issued and posted record correction patterns are operational.
- Documents, Portal, Mobile, BIM, Search and Reporting use source-versioned projections/references.
- Outbox, inbox, idempotency, dead-letter and reconciliation are production-ready.
- Retention, legal hold, backup, restore and disaster-recovery procedures are certified.
- Migration, import and merge tooling is auditable and resumable.
- No unresolved critical data-loss, integrity, isolation or recoverability defect remains.

50.2 Required architecture deliverables

Deliverable	Required content
Domain Ownership Registry	Every entity, aggregate and fact with owner and steward.
Logical ERD Pack	Foundation and per-domain relationship diagrams.
Physical Schema Catalogue	Tables, columns, keys, constraints, indexes and partitions.
Data Dictionary	Definitions, classifications, lineage and retention.
Reference Contract Registry	Typed relationships and exact-version policies.
Migration Catalogue	Schema versions, scripts, checksums and deployment order.
Retention Matrix	Record type, trigger, retention, hold and purge method.
Backup/DR Matrix	Tier, RPO, RTO, backup schedule and restore evidence.
Data Quality Catalogue	Rules, severity, ownership and correction path.
Test Certification Pack	Isolation, integrity, recovery, performance and migration evidence.

50.3 Required sign-off

Owner	Approval responsibility
Enterprise Architecture	Bounded contexts, references, consistency and deployment model.
Domain Owners	Aggregate ownership, invariants, lifecycle and correction rules.
Data Engineering	Warehouse lineage, rebuild, quality and reconciliation.
Security & Privacy	Isolation, classification, encryption, residency and sensitive access.
Platform / SRE	Capacity, observability, backup, restore and disaster recovery.
QA	Isolation, integrity, migration, recovery and performance certification.
Product Leadership	Open decisions, phased rollout and Definition of Done.

Final product position: ConstructionOS uses a domain-owned data architecture with one authoritative owner per business fact, explicit company ownership, local ACID transactions, stable cross-domain references, immutable revisions and postings, transactional events, permission-safe projections and rebuildable derived stores.